

HisabDo AI — Day 14

Budget Recommendation Methodology — Taha

1. Objective

Define an explainable MVP methodology for recommending category-level budgets using verified HisabDo historical spending and existing budget data. The recommendation engine must be deterministic, user-specific, and safe for insufficient data.

2. Taha — Exact Task Scope

1. Define budget recommendation methodology.
2. Analyze historical spending patterns.
3. Calculate recommended category budgets.
4. Compare current budget vs actual spending.
5. Identify overspending and underspending categories.
6. Define recommendation thresholds.
7. Validate recommendation logic with sample datasets.
8. Define evaluation criteria.
9. Document assumptions and limitations.
10. Validate final recommendation results.

3. MVP Methodology

1. Use the latest 3–6 complete months of verified expense history where available.
2. Group expenses by category and month.
3. Calculate each category's monthly spending average.
4. Use the historical monthly average as the initial recommended budget baseline.
5. Compare the baseline with the user's current budget when available.
6. Use configurable thresholds to classify overspending, underspending, or review-needed categories.
7. Do not use complex ML in the first MVP until the deterministic baseline has been validated.

4. Historical Spending Analysis

For every expense category, calculate:

- Total spending during the analysis period.
- Monthly spending values.
- Average monthly spending.
- Current-period spending where applicable.
- Existing budget amount where available.
- Difference between budget and actual spending.
- Budget utilization percentage when the budget is greater than zero.

5. Recommended Category Budget

MVP baseline: Recommended Budget = Average Monthly Category Spending over the verified analysis period.

Example: if Food spending is PKR 14,000, 16,000, and 15,000 across three months, the baseline recommendation is PKR 15,000.

6. Current Budget vs Actual Spending

- Budget Variance = Existing Budget – Actual/Historical Average Spending.
- Budget Utilization = $\text{Historical Average Spending} \div \text{Existing Budget} \times 100$, when existing budget > 0.
- If spending is above the existing budget, flag the category as overspending/review.
- If spending is below the existing budget, flag the category as underspending/within-budget review.
- Do not automatically treat underspending as a problem; it may simply indicate available budget headroom.

7. Recommendation Thresholds

Initial configurable MVP thresholds:

Overspending: Historical average > existing budget — Flag for budget review.

Significant overspending: Historical average $\geq$ existing budget $\times 1.20$ — High-priority review flag.

Within/near budget: Historical average is between 80% and 120% of existing budget — Normal review.

Underspending: Historical average < existing budget $\times 0.80$ — Show available budget headroom; do not force a reduction.

Insufficient history: Less than 2 complete months — Limited-data status; avoid strong recommendation.

Thresholds are configurable MVP starting points and must be validated on HisabDo sample/real test data before being treated as final production thresholds.

8. Sample Dataset Validation

Example test data:

- Food: 12,000 / 15,000 / 18,000; current budget 13,000.
- Transport: 8,000 / 7,000 / 6,000; current budget 10,000.
- Utilities: 5,000 / 5,500 / 5,200; no existing budget.
- Entertainment: only one month of data; current budget 6,000.

Expected validation: Food should be flagged for overspending review; Transport should show budget headroom; Utilities should receive a baseline recommendation if history is sufficient; Entertainment should be marked limited-data.

9. Evaluation Criteria

- Recommendation calculation matches the documented formula.
- Same input produces the same output.
- Categories are correctly grouped.
- Existing budgets are matched to the correct user and category.
- Overspending and underspending classifications follow configured thresholds.
- No division-by-zero errors occur.
- Insufficient data is handled safely.
- Recommendations do not use another user's data.
- AI explanation exactly reflects verified backend values.

10. Assumptions

- Only valid expense transactions are included.
- Transactions belong to the authenticated user.
- Category names are normalized before comparison.
- Analysis periods contain complete months where possible.
- Existing budget amounts are trusted backend values.
- The historical average is an MVP baseline, not a guaranteed future expense.

11. Limitations

- Historical averages may not capture seasonal expenses.
- One-time unusual expenses can increase the average.

- Short history may not represent normal behavior.
- Income changes are not enough by themselves to justify a category budget.
- Advanced ML/seasonality is outside the first MVP.

12. Final Validation Checklist

- ☐ Methodology finalized.
- ☐ Historical spending analysis verified.
- ☐ Category recommendations verified.
- ☐ Current budget comparison verified.
- ☐ Overspending/underspending rules verified.
- ☐ Thresholds documented and configurable.
- ☐ Sample dataset tested.
- ☐ Evaluation criteria documented.
- ☐ Assumptions and limitations documented.
- ☐ Final recommendation results validated.

13. Final Definition

Taha's Day 14 methodology establishes a simple, explainable and testable budget recommendation baseline. It uses verified historical category spending, compares it with existing budgets, applies configurable thresholds, validates results against sample data, and documents assumptions and limitations. The backend remains the source of truth for all recommendation values.